

Задание 19 - Кросс-компиляция

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- 1) Собрали конфигурацию под x86_64

```
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$ ./configure --prefix=$PWD/_install --static
Checking for gcc...
Building static library libz.a version 1.3.2 with gcc.
Checking for size_t... Yes.
Checking for off64_t... Yes.
Checking for fseeko... Yes.
Checking for strerror... Yes.
Checking for unistd.h... Yes.
Checking for stdarg.h... Yes.
Checking whether to use vs[n]printf() or s[n]printf()... using vs[n]printf().
Checking for vsnprintf() in stdio.h... Yes.
Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
Checking for s390x build ... No.
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$
```

- 2) Собираем под arm

```
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$ ^C
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$ ^C
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$ make CC=arm-linux-gnueabi-hf-gcc

arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_F_
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -o mini
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$ file trees.o
trees.o: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), not stripped
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$
```

- 3) прописал make install, по факту перенесли в нашу папку _install

```
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$ make install
rm -f /home/komal/Task_19_etc/zlib-1.3.2/_install/lib/libz.a
cp libz.a /home/komal/Task_19_etc/zlib-1.3.2/_install/lib
chmod 644 /home/komal/Task_19_etc/zlib-1.3.2/_install/lib/libz.a
rm -f /home/komal/Task_19_etc/zlib-1.3.2/_install/share/man/man3/zlib.3
cp zlib.3 /home/komal/Task_19_etc/zlib-1.3.2/_install/share/man/man3
chmod 644 /home/komal/Task_19_etc/zlib-1.3.2/_install/share/man/man3/zlib.3
rm -f /home/komal/Task_19_etc/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
cp zlib.pc /home/komal/Task_19_etc/zlib-1.3.2/_install/lib/pkgconfig
chmod 644 /home/komal/Task_19_etc/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
rm -f /home/komal/Task_19_etc/zlib-1.3.2/_install/include/zlib.h /home/komal/Task_19_etc/zlib-1.3.2/_install/include/zconf.h
cp zlib.h zconf.h /home/komal/Task_19_etc/zlib-1.3.2/_install/include
chmod 644 /home/komal/Task_19_etc/zlib-1.3.2/_install/include/zlib.h /home/komal/Task_19_etc/zlib-1.3.2/_install/include/zconf.h
komal@komal-VirtualBox:~/Task_19_etc/zlib-1.3.2$
```

- 4) Запускаем конфигурацию, но с нашей zlib

```

komal@komal-VirtualBox:~/Task_19_etc/openssh-portable$ ./configure --host=arm-linux-gnueabi --with-zlib=$PWD/../zlib-1.3.2/_install --prefix=$PWD/_install
checking for arm-linux-gnueabi-cc... no
checking for arm-linux-gnueabi-gcc... arm-linux-gnueabi-gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... yes
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether arm-linux-gnueabi-gcc accepts -g... yes
checking for arm-linux-gnueabi-gcc option to enable C11 features... none needed
checking if arm-linux-gnueabi-gcc supports C99-style variadic macros... yes
checking build system type... x86_64-pc-linux-gnu
checking host system type... arm-unknown-linux-gnueabi
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking for strings.h... yes
checking for sys/stat.h... yes
checking for sys/types.h... yes
checking forunistd.h... yes
checking whether byte ordering is bigendian... no
checking for gawk... no
checking for mawk... mawk
checking how to run the C preprocessor... arm-linux-gnueabi-gcc -E
checking for arm-linux-gnueabi-ld... arm-linux-gnueabi-ld

```

5) zlib встала успешна

```

checking for libgen.h... (cached) yes
checking for getspnam... yes
checking for library containing basename... none required
checking for sqrt... no
checking for sqrt in -lm... yes
checking for zlib... yes
checking for zlib.h... yes
checking for deflate in -lz... yes
checking for possibly buggy zlib... configure: WARNING: cross compiling: not checking zlib version
checking for strcasecmp... yes
checking for utimes... yes
checking for bsd/libutil.h... no
checking for libutil.h... no

```

6) теперь займемся с openssl Конфигурируем openssl под x86_64

```

komal@komal-VirtualBox:~/Task_19_etc/openssl-3.0.15$ ./Configure linux-armv4 --prefix=$PWD/_install --cross-compile-prefix=arm-linux-gnueabi --no-asm
--static no-shared no-dso
Configuring OpenSSL version 3.0.15 for target linux-armv4
Using os-specific seed configuration
Created configdata.pm
Running configdata.pm
Created Makefile.in
Created Makefile
Created include/openssl/configuration.h

*****
***      OpenSSL has been successfully configured      ***
***                                                    ***
***  If you encounter a problem while building, please open an ***
***  issue on GitHub <https://github.com/openssl/openssl/issues> ***
***  and include the output from the following command: ***
***                                                    ***
***      perl configdata.pm --dump ***
***                                                    ***
***  (If you are new to OpenSSL, you might want to consult the ***
***  'Troubleshooting' section in the INSTALL.md file first) ***
***                                                    ***
*****
komal@komal-VirtualBox:~/Task_19_etc/openssl-3.0.15$

```

7) Дальше собираем под arm

```

komal@komal-VirtualBox:~/Task_19_etc/openssl-3.0.15$ make CC=arm-linux-gnueabi-gcc AR=arm-linux-gnueabi-ar RANLIB=arm-linux-gnueabi-ranlib
./crypto/srp/libcrypto-lib-srp_lib.o
./crypto/srp/libcrypto-lib-srp_vfy.o
./crypto/libcrypto-lib-threads_lib.o
./crypto/libcrypto-lib-ctype.o
komal@komal-VirtualBox:~/Task_19_etc/openssl-3.0.15$ file ./crypto/libcrypto-lib-context.o
./crypto/libcrypto-lib-context.o: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), not stripped
komal@komal-VirtualBox:~/Task_19_etc/openssl-3.0.15$ s

```

8) Прописываю make install, чтобы все перенеслось в нашу папку _install

```

koma@koma-VirtualBox:~/Task_19_etc/openssl-3.0.15$ make install CC=arm-linux-gnueabi-gcc AR=arm-linux-gnueabi-ar RANLIB=arm-linux-gnueabi-ranl
ib
'make' depend && 'make' _build_libs
make[1]: Entering directory '/home/koma/Task_19_etc/openssl-3.0.15'
make[1]: Leaving directory '/home/koma/Task_19_etc/openssl-3.0.15'
make[1]: Entering directory '/home/koma/Task_19_etc/openssl-3.0.15'
make[1]: Leaving directory '/home/koma/Task_19_etc/openssl-3.0.15'
created directory '/home/koma/Task_19_etc/openssl-3.0.15/_install'
created directory '/home/koma/Task_19_etc/openssl-3.0.15/_install/lib'
*** Installing runtime libraries
*** Installing development files
created directory '/home/koma/Task_19_etc/openssl-3.0.15/_install/include'
created directory '/home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl'
install ./include/openssl/aes.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/aes.h
install ./include/openssl/asn1.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/asn1.h
install ./include/openssl/asn1_mac.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/asn1_mac.h
install ./include/openssl/asn1err.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/asn1err.h
install ./include/openssl/asn1t.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/asn1t.h
install ./include/openssl/async.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/async.h
install ./include/openssl/asyncerr.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/asyncerr.h
install ./include/openssl/bio.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/bio.h
install ./include/openssl/bioerr.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/bioerr.h
install ./include/openssl/blowfish.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/blowfish.h
install ./include/openssl/bn.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/bn.h
install ./include/openssl/bnerr.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/bnerr.h
install ./include/openssl/buffer.h -> /home/koma/Task_19_etc/openssl-3.0.15/_install/include/openssl/buffer.h

```

9) Запускаем конфигурацию, но с нашей zlib и openssl

```

koma@koma-VirtualBox:~/Task_19_etc/openssl-portable$
koma@koma-VirtualBox:~/Task_19_etc/openssl-portable$ ./configure --host=arm-linux-gnueabi --with-zlib=$PWD/../zlib-1.3.2/_install --with-ssl-dir=$
PWD/../openssl-3.0.15/_install --prefix=$PWD/_install
checking for arm-linux-gnueabi-gcc... no
checking for arm-linux-gnueabi-gcc... arm-linux-gnueabi-gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... yes
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether arm-linux-gnueabi-gcc accepts -g... yes
checking for arm-linux-gnueabi-gcc option to enable C11 features... none needed
checking if arm-linux-gnueabi-gcc supports C99-style variadic macros... yes
checking build system type... x86_64-pc-linux-gnu
checking host system type... arm-unknown-linux-gnueabi
checking for stdio.h... yes
checking for stdlib.h... yes
checking for string.h... yes

```

10) И видим что ssl встал как надо

```

checking whether snprintf can declare const char *fmt... yes
checking whether system supports SO_PEERCRED getsockopt... yes
checking if openpty correctly handles controlling tty... cross-compiling, assuming yes
checking whether AI_NUMERICSERV is declared... yes
checking if SA_RESTARTed signals interrupt select()... configure: WARNING: cross compiling: assuming yes
checking for getpgrp... yes
checking if getpgrp accepts zero args... yes
checking for openssl... /usr/bin/openssl
checking for openssl/opensslv.h... yes
checking OpenSSL header version... configure: WARNING: cross compiling: not checking
checking for OpenSSL_version... yes
checking for OpenSSL_version_num... yes
checking OpenSSL library version... configure: WARNING: cross compiling: not checking
checking whether OpenSSL's headers match the library... configure: WARNING: cross compiling: not checking
checking if programs using OpenSSL functions will link... yes
checking for BN_is_prime_ex... yes
checking for DES_crypt... yes
checking for EVP_DigestSign... yes

```

11) Далее собираем наши главные папки bin и lib где будут наши библиотеки


```

komal@komal-VirtualBox:~/Task_19_etc/openssh-portable$ make
Makefile:769: warning: ignoring prerequisites on suffix rule definition
conffile=echo sshd_config.out | sed 's/.out$//'; \
/usr/bin/sed -e 's|/etc/ssh/ssh_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_config|' -e 's|/etc/ssh/ssh_known_hosts|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_known_hosts|' -e 's|/etc/ssh/sshd_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshd_config|' -e 's|/usr/libexec|/home/komal/Task_19_etc/openssh-portable/_install/libexec|' -e 's|/etc/shosts.equiv|/home/komal/Task_19_etc/openssh-portable/_install/etc/shosts.equiv|' -e 's|/etc/ssh/ssh_host_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_key|' -e 's|/etc/ssh/ssh_host_ecdsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ecdsa_key|' -e 's|/etc/ssh/ssh_host_rsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_rsa_key|' -e 's|/etc/ssh/ssh_host_ed25519_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ed25519_key|' -e 's|/var/run/sshd.pid|/var/run/sshd.pid|' -e 's|/etc/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssl/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssh/sshrd|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshrd|' -e 's|/usr/X11R6/bin/xaauth|/usr/bin/xaauth|' -e 's|/var/empty|/var/empty|' -e 's|/usr/bin:/bin:/usr/sbin:/sbin|/usr/bin:/bin:/usr/sbin:/sbin|' /home/komal/Task_19_etc/openssh-portable/_install/bin|' ./${conffile} > sshd_config.out
conffile=echo sshd_config.out | sed 's/.out$//'; \
/usr/bin/sed -e 's|/etc/ssh/ssh_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_config|' -e 's|/etc/ssh/ssh_known_hosts|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_known_hosts|' -e 's|/etc/ssh/sshd_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshd_config|' -e 's|/usr/libexec|/home/komal/Task_19_etc/openssh-portable/_install/libexec|' -e 's|/etc/shosts.equiv|/home/komal/Task_19_etc/openssh-portable/_install/etc/shosts.equiv|' -e 's|/etc/ssh/ssh_host_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_key|' -e 's|/etc/ssh/ssh_host_ecdsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ecdsa_key|' -e 's|/etc/ssh/ssh_host_rsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_rsa_key|' -e 's|/etc/ssh/ssh_host_ed25519_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ed25519_key|' -e 's|/var/run/sshd.pid|/var/run/sshd.pid|' -e 's|/etc/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssl/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssh/sshrd|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshrd|' -e 's|/usr/X11R6/bin/xaauth|/usr/bin/xaauth|' -e 's|/var/empty|/var/empty|' -e 's|/usr/bin:/bin:/usr/sbin:/sbin|/usr/bin:/bin:/usr/sbin:/sbin|' /home/komal/Task_19_etc/openssh-portable/_install/bin|' ./${conffile} > sshd_config.out
conffile=echo moduli.out | sed 's/.out$//'; \
/usr/bin/sed -e 's|/etc/ssh/ssh_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_config|' -e 's|/etc/ssh/ssh_known_hosts|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_known_hosts|' -e 's|/etc/ssh/sshd_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshd_config|' -e 's|/usr/libexec|/home/komal/Task_19_etc/openssh-portable/_install/libexec|' -e 's|/etc/shosts.equiv|/home/komal/Task_19_etc/openssh-portable/_install/etc/shosts.equiv|' -e 's|/etc/ssh/ssh_host_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_key|' -e 's|/etc/ssh/ssh_host_ecdsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ecdsa_key|' -e 's|/etc/ssh/ssh_host_rsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_rsa_key|' -e 's|/etc/ssh/ssh_host_ed25519_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ed25519_key|' -e 's|/var/run/sshd.pid|/var/run/sshd.pid|' -e 's|/etc/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssl/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssh/sshrd|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshrd|' -e 's|/usr/X11R6/bin/xaauth|/usr/bin/xaauth|' -e 's|/var/empty|/var/empty|' -e 's|/usr/bin:/bin:/usr/sbin:/sbin|/usr/bin:/bin:/usr/sbin:/sbin|' /home/komal/Task_19_etc/openssh-portable/_install/bin|' ./${conffile} > moduli.out
conffile=echo moduli.out | sed 's/.out$//'; \
/usr/bin/sed -e 's|/etc/ssh/ssh_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_config|' -e 's|/etc/ssh/ssh_known_hosts|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_known_hosts|' -e 's|/etc/ssh/sshd_config|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshd_config|' -e 's|/usr/libexec|/home/komal/Task_19_etc/openssh-portable/_install/libexec|' -e 's|/etc/shosts.equiv|/home/komal/Task_19_etc/openssh-portable/_install/etc/shosts.equiv|' -e 's|/etc/ssh/ssh_host_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_key|' -e 's|/etc/ssh/ssh_host_ecdsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ecdsa_key|' -e 's|/etc/ssh/ssh_host_rsa_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_rsa_key|' -e 's|/etc/ssh/ssh_host_ed25519_key|/home/komal/Task_19_etc/openssh-portable/_install/etc/ssh_host_ed25519_key|' -e 's|/var/run/sshd.pid|/var/run/sshd.pid|' -e 's|/etc/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssl/moduli|/home/komal/Task_19_etc/openssh-portable/_install/etc/moduli|' -e 's|/etc/ssh/sshrd|/home/komal/Task_19_etc/openssh-portable/_install/etc/sshrd|' -e 's|/usr/X11R6/bin/xaauth|/usr/bin/xaauth|' -e 's|/var/empty|/var/empty|' -e 's|/usr/bin:/bin:/usr/sbin:/sbin|/usr/bin:/bin:/usr/sbin:/sbin|' /home/komal/Task_19_etc/openssh-portable/_install/bin|' ./${conffile} > moduli.out

```

- 12) Далее копируем в нашу корневую файловую систему собранные папки для ssh и запускаем qemu

```

komal@komal-VirtualBox:~/Task_19$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kernel zImage -initrd initramfs_ssh.cpio.gz -dtb vexpress-v2p-c9.dtb -append "console=ttyAMA0 rdinit=/bin/ash" -nographic

```

```

~ #
~ # ls
bin          initramfs.cpio.gz  sbin
dev          linuxrc           usr
etc          root             var
~ # ssh
Couldn't open /dev/null: No such file or directory
~ # which ssh
/bin/ssh
~ # ls -la bin/ssh
-rwxrwxr-x  1 1000  1000  6674088 Jun 15 10:50 bin/ssh
~ # file /bin/ssh
/bin/ash: file: not found
~ #

```

- 13) Настроил пользователя root, в корневую файловую систему в папку etc добавил два файла: shadow, passwd

```

~ # ssh -V
OpenSSH_10.3p1, OpenSSL 3.0.15 3 Sep 2024
~ #

```